BioPathNet++ for Autoimmune Diseases: Path-Based Transfer Learning for Rare Disease Drug Repurposing

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Abstract—Rare autoimmune diseases have few approved drugs, so computational repurposing must rank drugs for diseases whose treatments were never seen in training (zero-shot). We study this setting with BioPathNet++, a BioPathNet-style path-based graph neural network with an added channel of four leakage-free priors, on the PrimeKG knowledge graph. Our protocol holds out each test disease together with its related diseases and ranks all 7,957 drugs. On 68 unseen test diseases, a known indication is in the top 10 for 76% of diseases (random: 0.8%), and the mean reciprocal rank exceeds that of the best baseline (0.501 vs. 0.392; $p = 0.003$). With scarce training data, autoimmune source diseases transfer better than label-matched non-autoimmune ones (0.290 vs. 0.126; $p = 0.016$); with the full graph they do not help and even lower accuracy. Path reasoning carries the accuracy; the prior channel mainly adds stability. The output is a ranked list of hypotheses for expert review, not treatment advice.

Index Terms—drug repurposing, transfer learning, zero-shot learning, knowledge graphs, graph neural networks, autoimmune diseases

I. Introduction

Drug repurposing finds new uses for approved drugs, whose safety is already known, and is faster and cheaper than de novo development [1]. The need is greatest for rare diseases: only 5–7% of the more than 7,000 rare diseases have an FDA-approved drug [2]. Autoimmune disorders affected 10.2% of a 22-million-person UK cohort [3], yet primary biliary cholangitis (PBC), systemic sclerosis (SSc), Sjögren syndrome, autoimmune hepatitis (AIH), myasthenia gravis (MG) and Crohn disease have few disease-specific drugs and little curated data.

Knowledge graphs (KGs) link drugs, proteins, phenotypes and diseases, and link prediction on them ranks candidate drugs [4], [5]. For a disease without known treatments the task is zero-shot [2]: knowledge learned from diseases with known drugs must transfer to diseases whose drugs the model has never seen. Because autoimmune diseases share immune pathways, the best source of that knowledge might be the data-rich ones: rheumatoid arthritis (RA), lupus (SLE) and multiple sclerosis (MS). We ask three questions. Aim 1, source of transfer: must the knowledge come from related autoimmune diseases, or does broad pharmacology suffice? Aim 2, accuracy: does a path-based model beat simple repurposing rules on unseen diseases? Aim 3, components: what do path reasoning, the prior channel, the loss and the graph each contribute?

What is new relative to BioPathNet. BioPathNet++ uses the design of BioPathNet [6], a Neural Bellman–Ford network (NBFNet) [7] (Fig. 1); we do not claim a new architecture. Our contributions are (i) a strict-family zero-shot protocol that also hides each test disease's related diseases, motivated by a leakage audit of our own earlier pipeline; (ii) a controlled transfer study that compares autoimmune source diseases with label-matched non-autoimmune controls; (iii) a full-graph implementation whose memory is linear in the number of nodes, plus a prior channel that stabilises training; (iv) a head-to-head run of TxGNN's own code, showing that the winner depends on the candidate set; and (v) external checks of new candidates against trials, PubMed and later approvals.

II. Related Work

Biomedical KGs. Hetionet combined 29 public resources and ranked candidates with metapath features [4]. DRKG [8] and PrimeKG [9] followed; PrimeKG separates indications, contraindications and off-label uses for 17,080 diseases. Recent reviews trace KG-based repurposing from feature-based learning to graph neural networks [10], [11].

Embeddings and GNNs. KG embeddings [12]–[14] and relational GCNs [15] have been used for COVID-19 [16], [17], rare diseases [18] and general repurposing [19], [20]. Their reported gains depend strongly on how the dataset is built and evaluated [21], [22], and because they learn one vector per node from its training edges, they degrade for diseases without known drugs [2].

Zero-shot and path-based models. TxGNN handles such diseases with disease-similarity metric learning and reports 49.2% better zero-shot indication prediction than eight other methods [2]. NBFNet scores a (disease, drug) pair by Bellman–Ford message passing that starts at the disease, which makes it inductive and explainable by paths [7]; BioPathNet adapts it to biomedical KGs and evaluates it on TxGNN's disease-area splits [6]. Recent rare-disease studies instead build disease-specific graphs and explainable GNNs [23], [24].

Leakage. Redundant triples and loose splits overestimate repurposing accuracy [25], [26], and under heavy class imbalance AUROC can look high for weak rankings [27]. Table I positions BioPathNet++: it reuses

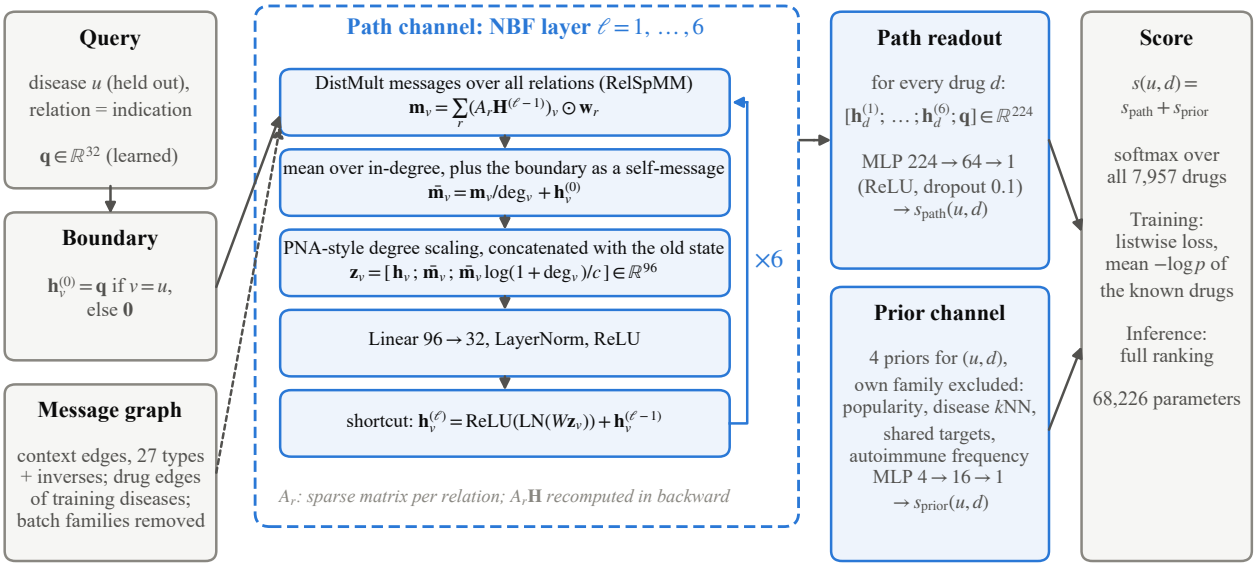


Fig. 1. BioPathNet++ architecture. Six NBF layers pass DistMult messages from the query disease u over a graph without the drug edges of u's family; a path MLP and a prior MLP (four leakage-free priors) give two scores whose sum ranks all 7,957 drugs.

BioPathNet's model and adds a stricter hold-out unit and a controlled transfer study.

TABLE I
Related approaches. KGE: KG embeddings. Induct.: scores diseases with no known drugs. Hold-out: unit removed for zero-shot testing (area: disease area). Transfer: controlled study of the training source.

Method	Induct.	Explains	Hold-out	Transfer
KGE [12]–[14]	no	no	–	no
R-GCN [15]	no	no	–	no
TxGNN [2]	yes	GraphMask	area	no
NBFNet [7]	yes	paths	–	no
BioPathNet [6]	yes	paths	area	no
BioPathNet++	yes	paths	disease+family	controlled

III. Data and Protocol

Knowledge graph and labels. We use PrimeKG [9]: 129,375 nodes (7,957 drugs, 17,080 diseases), 4,007,433 edges of 27 non-therapeutic relation types and 42,631 drug–disease pairs of three types: indication (the drug is labelled for the disease), off-label use and contraindication. We call indications and off-label uses known drugs, and use "approved" only for regulatory approvals that we checked ourselves. Validation and test positives are indications only. The rare targets have so few that their labels are indications plus off-label uses: PBC 1+0, SSc 3+2, Sjögren 2+10, AIH 0+1, MG 4+3, Crohn 16+8. Contraindications are never positives. With so few labels, the targets are case studies; statistical claims rest on larger held-out sets.

Preprocessing. PrimeKG's file lists every edge in both directions (8,100,498 rows). We keep one canonical direction per relation, drop duplicates, and add an explicit inverse for every relation (60 relation types in all). Drug–disease pairs are kept in a separate table, and each split's message graph receives only those of its training diseases. The path channel uses no node features or text embeddings: nodes differ only by their edges (bpp/build_primekg.py).

Leakage-free zero-shot protocol. A held-out disease is removed with its family: its disease–disease neighbours and, for targets, the subtypes that TxGNN's name-based grouping assigns to it. Without this rule, all 16 Crohn indications stayed reachable through "Crohn's colitis". No drug edge of a held-out family enters the message graph, and each training batch also hides its own diseases' drug edges, so training mimics zero-shot inference. An earlier version of our pipeline broke these rules: test pairs were graph edges, and a degree score with no learning reached AUROC 0.972. The six target families and 68 random validation diseases are held out; 1,206 diseases with indications remain for training.

Transfer conditions (Aim 1). The test set E holds the 12 diseases with indications in TxGNN's autoimmune area (descendants of DOID:417 [2]) that are neither targets nor RA/SLE/MS relatives. E and the targets are held out in every condition: A trains on the RA/SLE/MS families only (11 diseases, 161 indications); A′ on three random sets of 11 non-autoimmune diseases, each matched to A disease by disease on number of indications; D on all 1,194 non-autoimmune diseases; C_E on all 1,185 diseases except E. A vs. A′ asks whether autoimmune data helps when data is scarce, C_E vs. D whether it still helps at scale.

IV. Method

Path channel. For a query disease u, node states start from an indicator, $h_v^{(0)} = \mathbb{1}[v = u]\, q$, with a learned query

TABLE II
Metrics, per disease over all 7,957 drugs, averaged over diseases. Random: value for a random ranking.

Metric	Meaning	Random
Any@10	share of diseases with a known drug in the top 10	0.8%
H@k	share of known drugs in the top k (Hits@k)	k/7957
MRR	mean of 1/rank over known drugs (1 = all first)	≈0.001
AUROC	chance a known drug outranks a random other drug	0.5
AUPRC	area under precision–recall; rewards known drugs at the top	≈0.001

vector q, and pass through $L = 6$ layers

$$m_v^{(t)} = \frac{1}{\deg v} \sum_{(x,r,v)} h_x^{(t-1)} \odot w_r^{(t)} \; + \; \mathbb{1}[v = u]\, q, \tag{1}$$

$$h_v^{(t)} = h_v^{(t-1)} + \mathrm{ReLU}\big(\mathrm{LN}\big(W^{(t)}[h_v^{(t-1)};\, m_v^{(t)};\, a_v m_v^{(t)}]\big)\big), \tag{2}$$

with DistMult relation vectors w_r and a PNA-style degree scaler a_v [28]. An MLP over $[h_d^{(1)}; \ldots; h_d^{(L)}; q]$ scores every drug d in one pass (Fig. 1). The messages use one sparse matrix per relation and a custom autograd function that recomputes products in the backward pass, so memory grows with nodes, not edges.

Prior channel and training. A small MLP over four features adds a second score: drug popularity among training diseases, a disease-similarity kNN score ($k = 20$, TF–IDF over disease proteins and phenotypes), shared drug-target/disease-gene count, and frequency among autoimmune training diseases, each computed with the query's family excluded. The loss is the mean negative log-softmax probability of each disease's known drugs over all drugs. Hyperparameters were fixed in advance, not tuned.

Evaluation. Every drug is ranked, and ranks are filtered: a disease's other known drugs are removed before ranking each one [12]. Table II defines the metrics. MRR [29] rewards known drugs at the very top: on the test split, MRR 0.501 means a median known-drug rank of 2, with 43% ranked first. AUROC is high even for popularity (0.830), so it is never used alone [27]. The 68 validation diseases select the stopping epoch; 68 further test diseases are used neither for training nor for stopping. Of six non-neural baselines and three KG embeddings (TransE, DistMult, RotatE, trained on the same graph), the strongest is disease kNN, which transfers the drugs of the 20 most similar training diseases. CIs are bootstraps over diseases [30]; a difference is significant only if the paired CI excludes zero and a two-sided Wilcoxon test gives $p < 0.05$ [31].

Reproducibility. Data: PrimeKG (Harvard Dataverse, doi:10.7910/DVN/IXA7BM) and TxGNN's area files and code (GitHub, commit 0e0203c). Splits: validation seed 0, test seed 1, matched controls seeds 100–102, all stored as JSON. Model: dimension 32, 6 layers, readout MLP 224→64→1 (dropout 0.1), prior MLP 4→16→1; 68,226 parameters. Training: Adam, learning rate 5×10^{-3}, 8 diseases per batch, gradient clipping 1.0, ≤20 epochs, patience 5–6 on validation MRR (transfer runs: ≤60 epochs, patience 10); seeds 0–2; one NVIDIA T4 (16 GB) per run, 8.5 s per step, 5.8 GB peak. Statistics: 1,000 bootstrap resamples (5,000 for paired tests). Code, splits, logs and the script that generates every number: https://github.com/SaiYatin/Capstone_Drug_Repurposing_Project (branch bpp-v3).

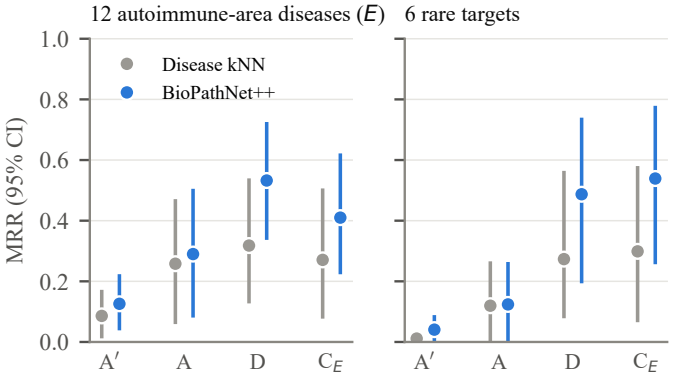


Fig. 2. Aim 1: MRR on held-out autoimmune diseases (E, left) and the targets (right) when training on A′, A, D or C_E (95% CIs).

V. Results

A. Aim 1: transfer to rare autoimmune diseases

Scarce data (Fig. 2). Trained only on the RA/SLE/MS families (A), BioPathNet++ reaches MRR 0.290 on E, against 0.126 with label-matched non-autoimmune diseases (A′): a significant difference of +0.164 (CI +0.037 to +0.316; p=0.016; better on 9 of 12), which also holds for final instead of early-stopped checkpoints (p=0.021). The rare targets point the same way (0.124 vs. 0.041; p=0.062, $n = 6$), and so does disease kNN (0.258 vs. 0.086). What transfers is a drug-class prior: the A-trained model ranks the same corticosteroids first for every target, a sensible first guess but not disease-specific reasoning. With paths only, A and A′ both stay near the floor (0.007 vs. 0.010), so the advantage comes from the prior channel. Full scale. Training on all non-autoimmune diseases (D) reaches 0.532 on E; adding the autoimmune diseases (C_E, 3 seeds) lowers it to 0.410 (p=0.010; same validation set), and disease kNN drops too (0.318 to 0.271). On the targets, removing all autoimmune diseases leaves MRR unchanged (0.487 vs. 0.475; p=0.812). The best source thus depends on data size: related autoimmune diseases when data is scarce, the broad pharmacological graph at full scale.

B. Aim 2: accuracy on unseen diseases

On the selection-free test split (Table III), a known indication is in the top 10 for 76% of diseases, against 57% for disease kNN, 19% for popularity and 0.8% by chance. MRR is 0.501 vs. 0.392 (difference +0.109, CI +0.044 to +0.180; p=0.003). On the 68 validation diseases the gain

TABLE III
Aim 2: ranking of all 7,957 drugs. Any@10: diseases with a known indication in the top 10 (random 0.8%); H@k: known drugs in the top k; random AUPRC ≈0.001. Test: seed means (3 seeds; KG embeddings 3).

Set	Method	Any@10	H@1	H@10	H@50	AUROC	$\text{AUROC}_{\text{hard}}$	$\text{AUROC}_{\text{deg}}$	AUPRC	MRR	Mean rank
Test (68, never selected)	Random	1%	0.000	0.005	0.006	0.453	0.471	0.500	0.002	0.001	4,349
	Popularity	19%	0.042	0.070	0.267	0.885	0.562	0.812	0.084	0.062	918
	TransE	27%	0.019	0.061	0.146	0.880	0.674	0.773	0.050	0.037	952
	DistMult	22%	0.025	0.089	0.193	0.852	0.638	0.732	0.060	0.047	1,175
	RotatE	27%	0.052	0.094	0.209	0.878	0.723	0.785	0.089	0.071	971
	Disease kNN	57%	0.347	0.459	0.583	0.907	0.721	0.859	0.429	0.392	742
	BioPathNet++	76%	0.435	0.637	0.727	0.959	0.862	0.924	0.556	0.501	329
Validation (68)	Random	1%	0.000	0.002	0.004	0.471	0.488	0.490	0.002	0.001	4,205
	Popularity	15%	0.034	0.049	0.188	0.830	0.476	0.760	0.069	0.046	1,355
	Disease kNN	44%	0.239	0.350	0.490	0.860	0.661	0.822	0.313	0.281	1,118
	BioPathNet++	56%	0.353	0.456	0.601	0.938	0.799	0.886	0.424	0.394	495
Rare targets (6)	Random	0%	0.000	0.000	0.000	0.476	0.488	0.389	0.002	0.001	4,165
	Popularity	50%	0.062	0.148	0.162	0.905	0.460	0.766	0.116	0.099	755
	Disease kNN	67%	0.243	0.424	0.725	0.962	0.778	0.917	0.407	0.296	299
	BioPathNet++	83%	0.396	0.648	0.695	0.982	0.892	0.954	0.594	0.475	141

TABLE IV
Evidence for unlabelled top-20 candidates (6 targets) and two reference sets (Ours n=86; Pop., Random n=120; grades as in the text).

Evidence	Ours	Pop.	Random
Trial names drug, A–C	17%	14%	0%
Trial names drug, A–B	15%	12%	0%
Trial tests drug, A–C	13%	8%	0%
Trial tests drug, A–B	12%	8%	0%
PubMed co-mention	63%	72%	18%
PubMed: benefit	15%	10%	0%
PubMed: harm/no benefit	5%	12%	0%
Target among disease genes	17%	16%	3%

over kNN is +0.113 ($p<0.001$). BioPathNet++ therefore outperforms every baseline evaluated on the same leakage-free split. On the six targets it is ahead for 4 of 6, not significant with $n = 6$.

C. Aim 3: what each component contributes

Path reasoning carries the signal. On the 68 validation diseases, paths alone reach MRR 0.405, tied with the full model's 0.394 ($p=0.148$); the prior channel alone reaches 0.306. These validation values are not comparable with the test-split MRR of 0.501: that is the full model on 68 other diseases, which are easier for every method (kNN 0.392 on test vs. 0.281 on validation). The prior channel does not raise the mean but reduces seed variance on the targets (sd 0.017 vs. 0.087). Self-adversarial BCE is not significantly worse than the listwise loss (0.378 vs. 0.394; $p=0.344$), and dropping tissue-expression and drug–drug edges removes 72% of edges at unchanged accuracy ($p=0.299$).

D. Case studies and validation

Known drugs. Ursodeoxycholic acid ranks first for PBC in every seed, and the four labelled cholinesterase inhibitors hold ranks 1–4 for MG. AIH scores near zero only because PrimeKG lists one off-label drug for it, while the model's top choice, prednisolone, is standard AIH therapy (comparator in NCT00838214). Unlabelled candidates (Table IV). We searched ClinicalTrials.gov for each candidate and for two reference sets (popularity top 20; 20 random drugs per target). The drug must be a named intervention, matched by whole name (A: phase ≥3, B: phase 2, C: phase 1/observational). A trial tests the drug only if its title names it or the drug is in some but not all arms, which excludes background therapy, premedication and transplant conditioning. 13% of the model's candidates have such a trial (9 at phase ≥3), against 8% for popularity and 0% for random drugs; examples are amifampridine for MG (NCT03304054), ciprofloxacin for Crohn (NCT00609973) and N-acetylcysteine for SSc (NCT00428883), all checked by hand. A trial is not proof of benefit: amifampridine helped in MuSK-antibody MG [32] but not in acetylcholine-receptor-antibody MG [33]. Ustekinumab for Crohn counts as unlabelled only because PrimeKG misses its 2016 approval. PubMed titles were classified with AI assistance and every benefit or harm label was re-checked against the abstract. PubMed shows benefit for 15% of the model's candidates (popularity 10%, random 0%) and harm or no benefit for 5% (popularity 12%): three corticosteroid pairs and N-acetylcysteine for SSc, whereas popularity's are mostly antibiotics that can worsen MG or induce hepatitis. Temporal check. Six drugs approved after the PrimeKG snapshot (risankizumab, upadacitinib, guselkumab for Crohn; ravulizumab, rozanolixizumab for MG; elafibranor for PBC) rank at a median of 1,104 of 7,957; 5 of 6 beat chance, not significant ($p=0.109$). Explanations and bias. Gradient-based path extraction [7] shows, for example, that amifampridine reaches MG through its indication for Lambert–Eaton syndrome, a related neuromuscular-junction disease, and exposes weak links such as ciprofloxacin reaching Crohn through "mouth disease". Corticosteroids fill 41% and antibacterials 21% of top-20 slots, so the lists need expert filtering.

VI. Discussion and Limitations

Our MRR values lie within the range BioPathNet reports on TxGNN's disease areas (about 0.2–0.8) [6], but splits differ, so we ran TxGNN's own code on its autoimmune split, with our model trained only on edges that TxGNN also trains on (3 seeds, 13 diseases). Ranking all drugs, BioPathNet++ is significantly better (MRR 0.273 vs. 0.048, p=0.040): only 23% of TxGNN's top-100 drugs have any indication, against 99% for ours. Ranking only drugs with indications, as TxGNN does, no difference is significant (Recall@100 0.614 vs. 0.536; AUROC 0.777 vs. 0.901). BioPathNet was not re-run. Limitations: (i) the targets have 1–24 labelled drugs, so their metrics are noisy and can be lower bounds (AIH); (ii) the validation diseases also choose the stopping epoch (0.002–0.016 MRR of optimism), which the test split avoids; (iii) E has only 12 diseases, and 11-disease models are unstable across seeds; (iv) literature labels have not been reviewed by a clinician; (v) corticosteroids and antibiotics dominate the lists.

VII. Conclusion

Path-based zero-shot transfer works for rare autoimmune diseases the model never sees. When training data is scarce, related autoimmune diseases are the most useful source; at full scale the broad pharmacological graph is the better source. BioPathNet++ outperforms every baseline evaluated on the same leakage-free split, recovers established therapies, and ranks later-approved drugs high but not significantly above chance. Path reasoning drives accuracy, and the prior channel adds stability.

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References

[1] S. Pushpakom, F. Iorio, P. A. Eyers, K. J. Escott, S. Hopper, A. Wells et al., "Drug repurposing: progress, challenges and recommendations," Nature Reviews Drug Discovery, vol. 18, no. 1, pp. 41–58, 2019.

[2] K. Huang, P. Chandak, Q. Wang, S. Havaldar, A. Vaid, J. Leskovec, G. N. Nadkarni, B. S. Glicksberg, N. Gehlenborg, and M. Zitnik, "A foundation model for clinician-centered drug repurposing," Nature Medicine, vol. 30, no. 12, pp. 3601–3613, 2024.

[3] N. Conrad, S. Misra, J. Y. Verbakel, G. Verbeke, G. Molenberghs et al., "Incidence, prevalence, and co-occurrence of autoimmune disorders over time and by age, sex, and socioeconomic status: a population-based cohort study of 22 million individuals in the UK," The Lancet, vol. 401, no. 10391, pp. 1878–1890, 2023.

[4] D. S. Himmelstein, A. Lizee, C. Hessler, L. Brueggeman, S. L. Chen, D. Hadley, A. Green, P. Khankhanian, and S. E. Baranzini, "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," eLife, vol. 6, p. e26726, 2017.

[5] S. Bonner, I. P. Barrett, C. Ye, R. Swiers, O. Engkvist, A. Bender, C. T. Hoyt, and W. L. Hamilton, "A review of biomedical datasets relating to drug discovery: a knowledge graph perspective," Briefings in Bioinformatics, vol. 23, no. 6, p. bbac404, 2022.

[6] E. Y. Hu, S. Oleshko, S. Firmani, H. G. Cheng, Z. Zhu, M. A. Wörheide, M. Arnold, M. Colomé-Tatché, J. Tang, S. Xhonneux, and A. Marsico, "Enhancing link prediction in biomedical knowledge graphs with BioPathNet," Nature Biomedical Engineering, vol. 10, no. 8, pp. 1747–1769, 2026.

[7] Z. Zhu, Z. Zhang, L.-P. Xhonneux, and J. Tang, "Neural Bellman-Ford networks: A general graph neural network framework for link prediction," in Advances in Neural Information Processing Systems (NeurIPS), 2021, arXiv:2106.06935.

[8] V. N. Ioannidis, X. Song, S. Manchanda, M. Li, X. Pan, D. Zheng, X. Ning, X. Zeng, and G. Karypis, "DRKG – drug repurposing knowledge graph for COVID-19," https://github.com/gnn4dr/DRKG, 2020.

[9] P. Chandak, K. Huang, and M. Zitnik, "Building a knowledge graph to enable precision medicine," Scientific Data, vol. 10, p. 67, 2023.

[10] S. Wei, C. Sasi, J. Piepenbrock, M. A. Huynen, and P. A. C. 't Hoen, "The use of knowledge graphs for drug repurposing: From classical machine learning algorithms to graph neural networks," Computers in Biology and Medicine, vol. 196, p. 110873, 2025.

[11] A. Kumar, S. Bhandary, S. G. Hegde, and J. Chatterjee, "Knowledge graph applications and multi-relation learning for drug repurposing: A scoping review," Computational Biology and Chemistry, vol. 115, p. 108364, 2025.

[12] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in Advances in Neural Information Processing Systems (NeurIPS), 2013.

[13] B. Yang, W.-t. Yih, X. He, J. Gao, and L. Deng, "Embedding entities and relations for learning and inference in knowledge bases," in International Conference on Learning Representations (ICLR), 2015, arXiv:1412.6575.

[14] Z. Sun, Z.-H. Deng, J.-Y. Nie, and J. Tang, "RotatE: Knowledge graph embedding by relational rotation in complex space," in International Conference on Learning Representations (ICLR), 2019, arXiv:1902.10197.

[15] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in The Semantic Web (ESWC), ser. Lecture Notes in Computer Science, vol. 10843, 2018, pp. 593–607.

[16] R. Zhang, D. Hristovski, D. Schutte, A. Kastrin, M. Fiszman, and H. Kilicoglu, "Drug repurposing for COVID-19 via knowledge graph completion," Journal of Biomedical Informatics, vol. 115, p. 103696, 2021.

[17] D. Morselli Gysi, Í. do Valle, M. Zitnik, A. Ameli, X. Gan et al., "Network medicine framework for identifying drug-repurposing opportunities for COVID-19," Proceedings of the National Academy of Sciences, vol. 118, no. 19, p. e2025581118, 2021.

[18] D. N. Sosa, A. Derry, M. Guo, E. Wei, C. Brinton, and R. B. Altman, "A literature-based knowledge graph embedding method for identifying drug repurposing opportunities in rare diseases," Pacific Symposium on Biocomputing, vol. 25, pp. 463–474, 2020.

[19] Z. Gao, P. Ding, and R. Xu, "KG-Predict: A knowledge graph computational framework for drug repurposing," Journal of Biomedical Informatics, vol. 132, p. 104133, 2022.

[20] D. Bang, S. Lim, S. Lee, and S. Kim, "Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers," Nature Communications, vol. 14, p. 3570, 2023.

[21] S. Bonner, I. P. Barrett, C. Ye, R. Swiers, O. Engkvist, C. T. Hoyt, and W. L. Hamilton, "Understanding the performance of knowledge graph embeddings in drug discovery," Artificial Intelligence in the Life Sciences, vol. 2, p. 100036, 2022.

[22] M. Ali, M. Berrendorf, C. T. Hoyt, L. Vermue, M. Galkin, S. Sharifzadeh, A. Fischer, V. Tresp, and J. Lehmann, "Bringing light into the dark: A large-scale evaluation of knowledge graph embedding models under a unified framework," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 12, pp. 8825–8845, 2022.

[23] F. Papadimas, S. Svolou, K. Bougiatiotis, F. Aisopos, A. Krithara et al., "Identifying drug repurposing candidates for rare neuro-muscular disorders, using different AI methods on the literature knowledge graph," JAMIA Open, vol. 9, no. 2, 2026.
[24] P. Perdomo-Quinteiro, K. Wolstencroft, M. Roos, and N. Queralt-Rosiñach, "Knowledge graphs and explainable artificial intelligence (AI) for drug repurposing on rare diseases," Neurosymbolic Artificial Intelligence, vol. 2, 2026.
[25] G. Brière, T. Stosskopf, B. Loire, and A. Baudot, "Benchmarking the impact of data leakage on the performance of knowledge graph embedding models for biomedical link prediction," Bioinformatics, vol. 42, no. 8, p. btag608, 2026.
[26] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, p. 100804, 2023.
[27] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLoS ONE, vol. 10, no. 3, p. e0118432, 2015.
[28] G. Corso, L. Cavalleri, D. Beaini, P. Liò, and P. Veličković, "Principal neighbourhood aggregation for graph nets," in Advances in Neural Information Processing Systems (NeurIPS), 2020, arXiv:2004.05718.
[29] E. M. Voorhees, "The TREC-8 question answering track report," in Proceedings of the Eighth Text REtrieval Conference (TREC-8), 1999.
[30] B. Efron and R. J. Tibshirani, An Introduction to the Bootstrap. New York: Chapman & Hall, 1993.
[31] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bulletin, vol. 1, no. 6, pp. 80–83, 1945.
[32] S. Bonanno, M. B. Pasanisi, R. Frangiamore, L. Maggi, C. Antozzi, F. Andreetta, A. Campanella, G. Brenna, L. Cottini, and R. Mantegazza, "Amifampridine phosphate in the treatment of muscle-specific kinase myasthenia gravis: a phase IIb, randomized, double-blind, placebo-controlled, double crossover study," SAGE Open Medicine, vol. 6, p. 2050312118819013, 2018.
[33] L. Remijn-Nelissen, W. R. Bakker, T. van Gelder, D. J. A. R. Moes, A. M. Ruiter, Y. J. M. Campman, U. A. Badrising, C. S. M. Straathof, J. J. G. M. Verschuuren, and M. R. Tannemaat, "Efficacy and safety of amifampridine in myasthenia gravis: A randomized, double-blind, placebo-controlled crossover trial," Neurology, vol. 106, no. 8, p. e214715, 2026.